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Introduction to Video Analysis

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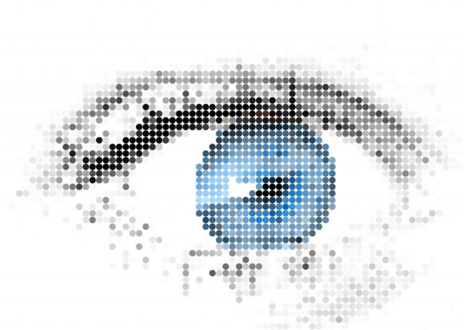
From Image to Video Analysis



- Computer vision aims to understand images
- Rapid growth over the last few years (due to deep learning)
  - ability to detect obstacles, segment images, or extract relevant context from a given scene
- **Video** is starting to garner more attention
  - Most applications center on images, with less focused on sequences of images

Video Analysis Definition

- It is not a new area
- 1878: Photographic experiment - Do horses ever lift all four feet off the ground during their gallop?
- Multiple cameras to obtain 24 still images at different times

the goal is to obtain position and time data from the frames of a video

What is a Video?

An ordered set of frames (images)

- Fixed-length (duration)
- Resolution (360p, ..., 1080p, ..., 4k)
    - Same resolution throughout the sequence
- Color (Grayscale or RGB)
- Frames amount (e.g. 3-5 frames/sec)

What is a Video?

When analyzing a video, we have a video stream or a video sequence

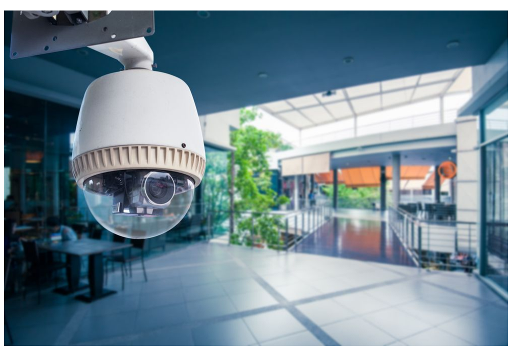
- Video stream (live image feed): we consider the current image and the previous ones.

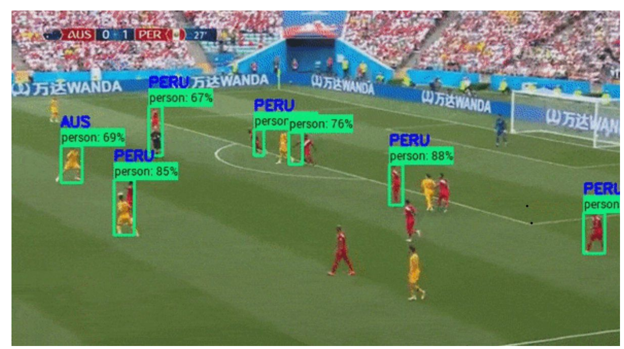
- Video sequence (fixed-length video): we have access to the full video, from the first image to the last.

Motivation

Why Video is important?

- Useful applications
  - Surveillance camera systems, sport analysis, etc.
- Implies many modalities
  - Audio, text, images, etc



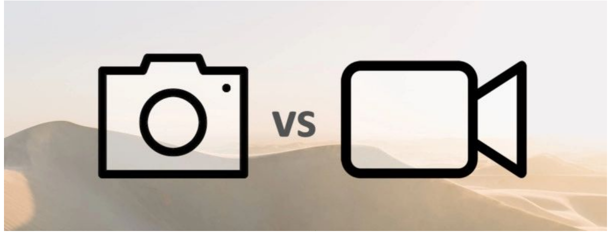


Motivation

Why Video Analysis is important?

- Automation
  - Most cameras only record, with little or not automation
- Explore the world
  - Robots, self-driving cars, etc. (detecting objects, location)
- Much to solve yet
  - Video analysis is challenging
  - Many unsolved problems

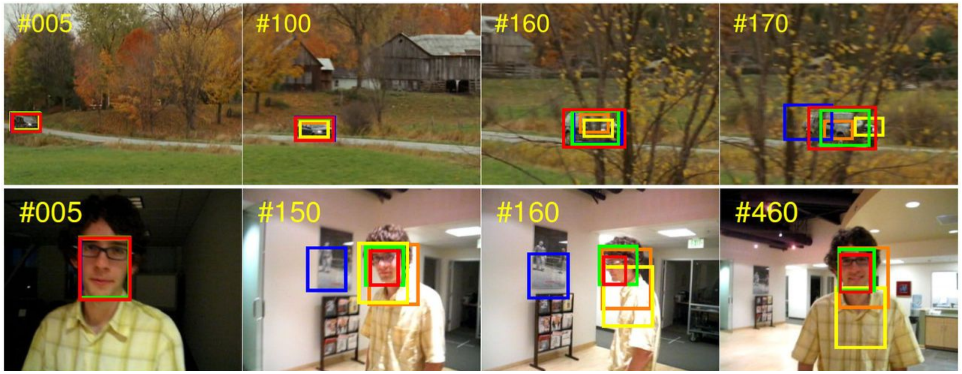
Image vs Video Analysis



- By definition, video is a sequence of images
  - Frames are spatially and temporally related
- Key difference: **Motion**
  - It’s a powerful feature to video understanding
- Multimodality
  - At least image and audio, it could be used to video analysis
- Size
  - Videos need more storage and processing capacity

Areas of Video Analysis

- Visual Object Tracking (VOT)
  - The task is to locate a certain object in all frames of a video, given only its location in the first frame
- Action Classification/Recognition
  - The goal is to analyze a video to identify the actions taking place in the video
  - Sometimes depends on object tracking

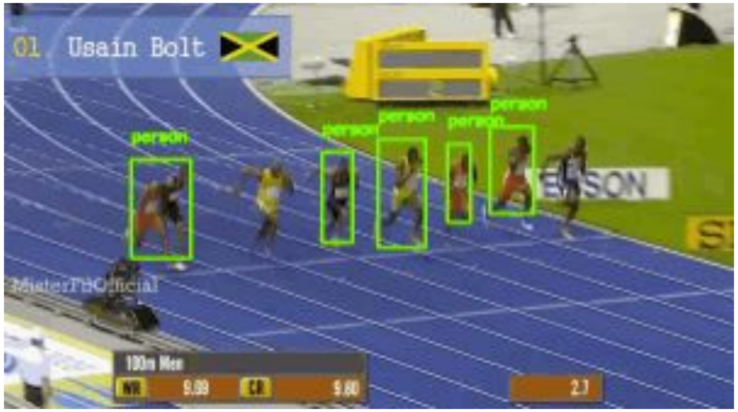


Visual Object Tracking (VOT)

- One of the main challenges in computer vision
- The task is to locate an object in all frames of a video
- **Condition**: the location of the object is given in the first frame
- Object refers to things or people

Visual Object Tracking (VOT)

- It is not necessary to know what the object is
- Only previous frames are used (stream approach)
- Commonly used for short-term tracking
- Multiple objects is possible (MOT)
- Example

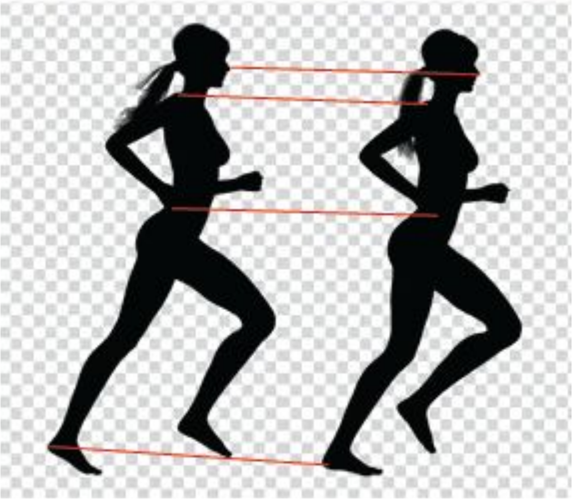


Challenges of VOT

- Computation load
    - Real applications
- Apparence
    - Changes over the time, due to dynamics, lighting, viewpoint, etc.
- Object Interaction
    - Occlusion and similarity with other objects
- Motion
    - Flow estimation

Optical Flow

- Optical flow estimation is key problem in video analysis
  - computing a pixel shift between two frames
  - it is handled as a correspondence problem



Optical Flow

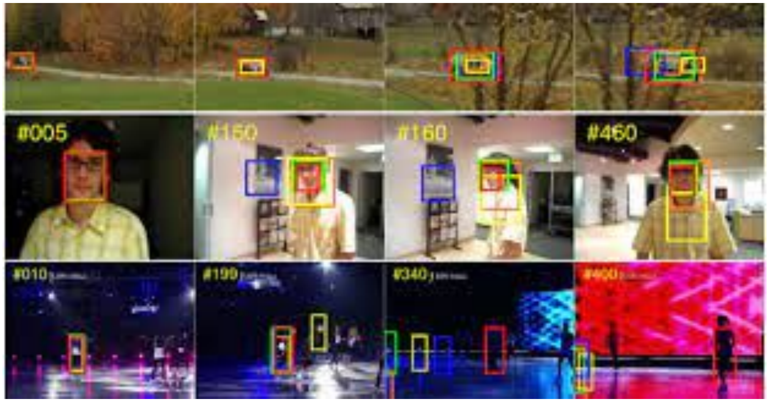
- It helps to understand the movement of pixels from one frame to another
- The output is a vector of movement between frame 1 and frame 2

Datasets

- Ground-true is difficult to obtain
- There are several data sets, but they are not suitable for training deep learning models.

VOT Challenge (Kristan, 2020)

- Open source implementations
- Evaluation framework
- Short videos (100 frames)
- Small amount of videos



Action Recognition

Human actions is the main content of videos

- Movies, TV shows, video surveillance, etc.

The objective is to detect an action or an event in a video.

Detecting actions is useful for:

- Smart surveillance (detection of abnormal situations)
- Video retrieval (search for a video with a specific query)
- Content Navigation (find a step of a recipe)

Action Recognition

Actions

- Small actions carried out for a common goal

Events

- An action could be called event
- Events can include multiple actions (different people or objects)

Tasks of Action Recognition

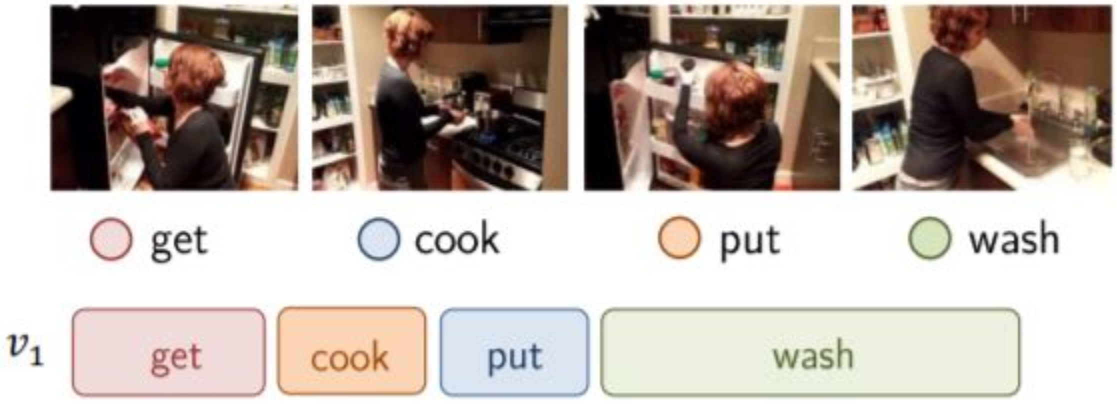
- Action classification
  - Predict the action label of a video

phoning *running* *walking* *ridinghorse*

- Action localization/detection
  - Find the spatial region and/or time interval of action in a video

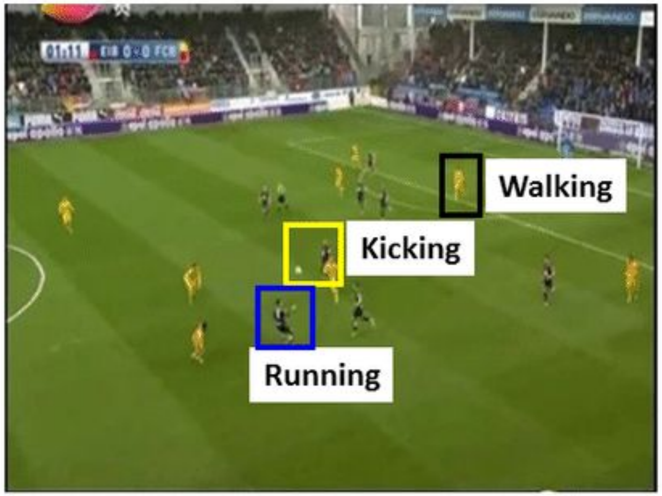
Complex Action Recognition Tasks

- Complex activities
  - Daily life activities implies several actions
  - It is a sequence of actions to a common goal
  - E.g. cooking, washing, etc.



Complex Action Recognition Tasks

- Multiple actions
    - Real scenarios contain multiple actions
    - Actions can be performed simultaneously
    - Actions can be performed by an human or object

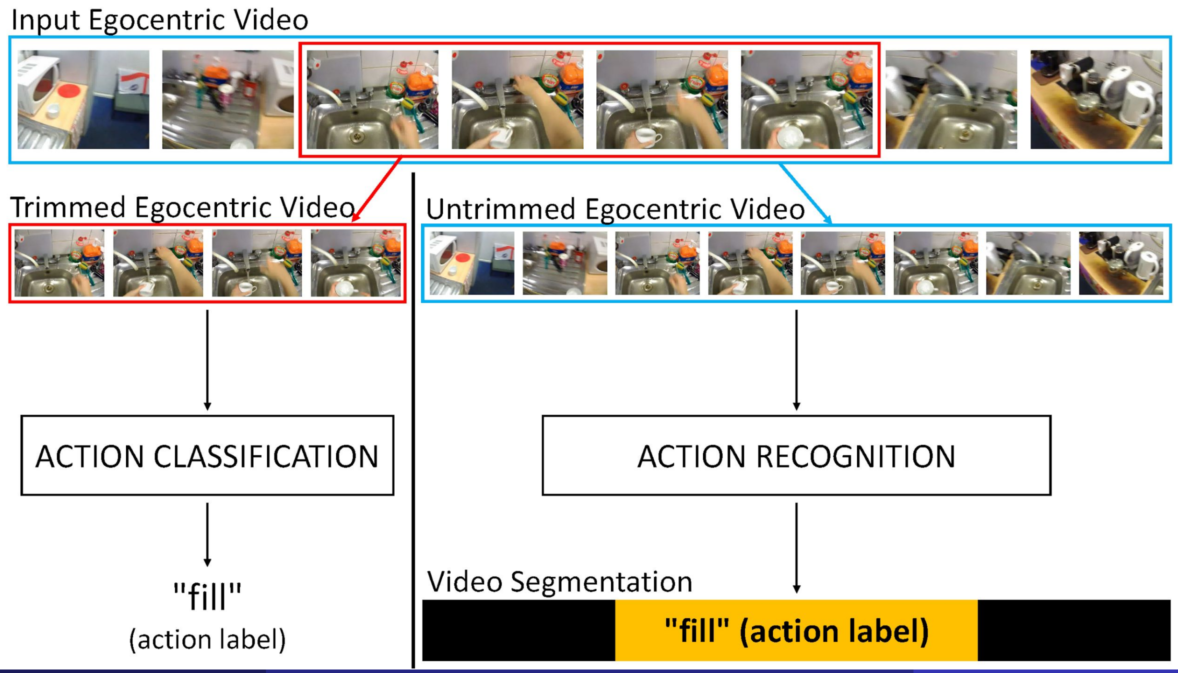


Complex Action Recognition Tasks

- Egocentric action recognition
  - Also know as first-person action recognition
  - Entails analyzing videos captured by a wearable camera
  - Useful for robotic applications

Trimmed and Untrimmed Videos

- Trimmed
  - Videos showing only one action
- Untrimmed
  - Videos contain additional information that may not be relevant
  - More realistic scenarios (e.g. web videos, security cameras, etc)



Challenges of Action Recognition

Basic human actions are easy to recognize

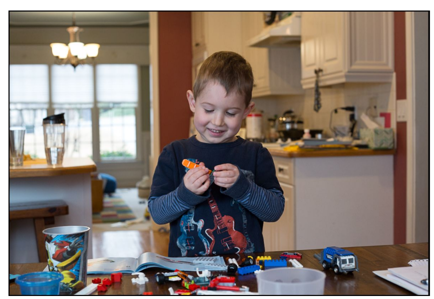
Boxing

Walking

Hand waving

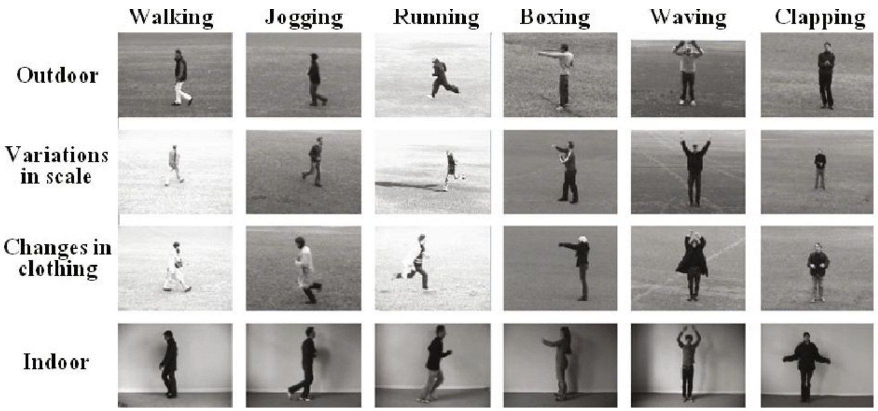
However, it is a challenging task due to problems:

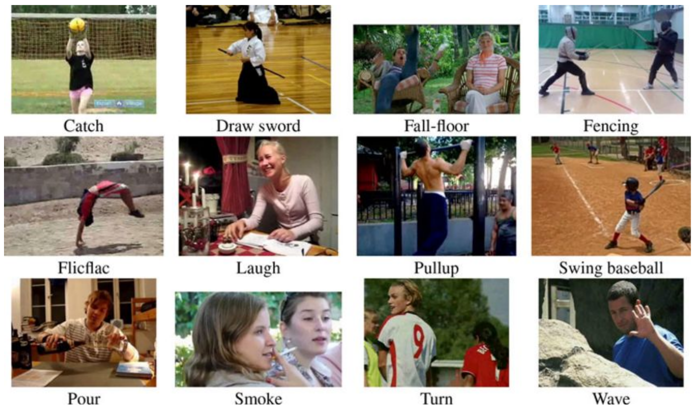
- background clutter
- partial occlusion
- changes in scale
- viewpoint
- lighting
- appearance



Basic Datasets

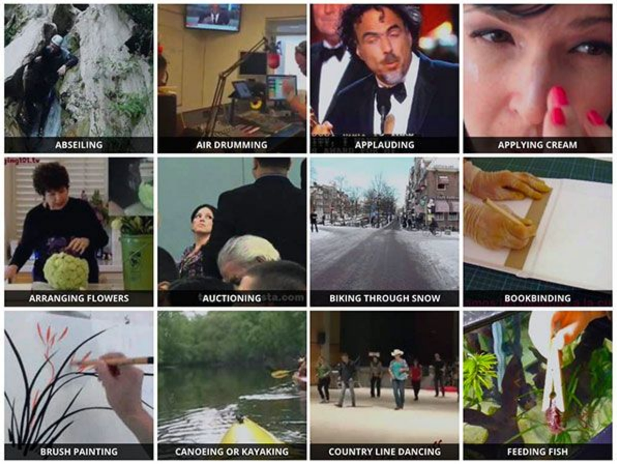
- KHT Actions (Schuldt, 2004)
  - 25 people
  - 6 actions
  - 4 scenes (inside, outside, scales, clothes)
  - 2,391 videos
- HMDB (Kuehne, 2011)
  - 51 actions
  - 101 videos per action
  - 6,849 videos
  - Video stabilization





Big Datasets

- UCF101 (Soomro, 2012)
  - 101 classes
  - 13,320 Youtube videos
  - 5 groups: human-object interaction, human-human interaction, body-motion, playing instruments, sports
- Kinetics (Kay, 2017)
  - 400 human action classes
  - 650,000 videos
  - Each action class has at least 400/600/700 videos
  - human-object interactions, human-human interactions
  - lasts around 10 seconds



Complex Datasets

- 20BN-something-something (Goyal, 2017)
    - V2: 220,847 videos, V1: 108.499 videos
    - 174 labels
    - labels are compound of interaction between objects
        - For example, "Putting [something] onto [something]"
    - there are 318,572 annotations involving 30,408 unique objects
    - reduced label noise

Trying to pour water into a glass, but missing so it spills next to it

Spinning a bracelet so it continues spinning

Pulling two ends of a rubber band so that it gets stretched

Complex Datasets

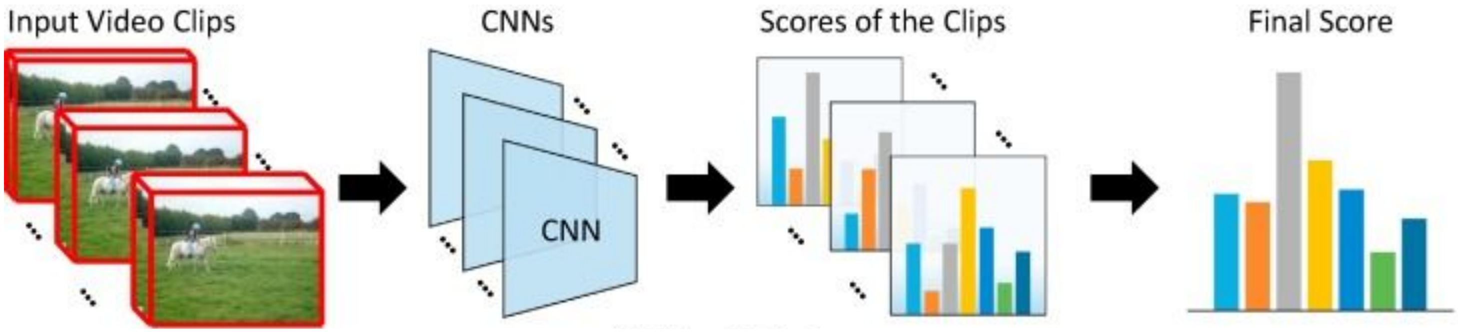
EPIC-KITCHENS (Damen, 2020)

- Egocentric video dataset
  - 32 individuals in 4 cities in different countries
  - 55 hours of video consisting of 11.5M frames
  - labelled for a total of 39.6K action segments
  - 454.2K object bounding boxes
- Challenges:
  - Action Recognition
  - Action Detection
  - Action Anticipation
  - Domain Adaptation for Action Recognition
  - Multi-Instance Retrieval

How DL is Used for Action Recognition

A simple approach is to use 2D CNN

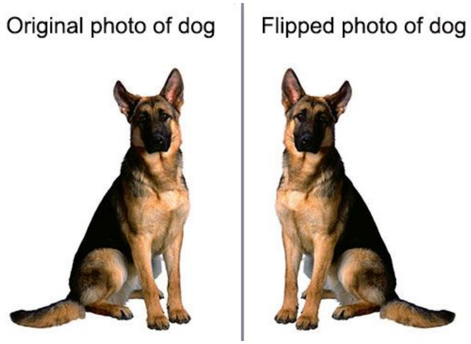
- Feed each frame to a CNN
- Predict at every frame
- Optional: use an intermediate layer to fuse information
- Optional: use pre-trained networks on images

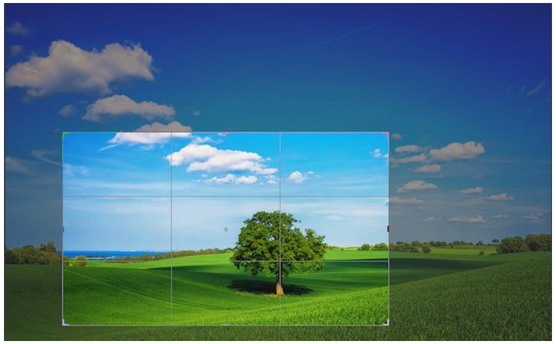


How DL is Used for Action Recognition

For DL usage, we need to preprocess video

- Treat as image
- Apply transformations





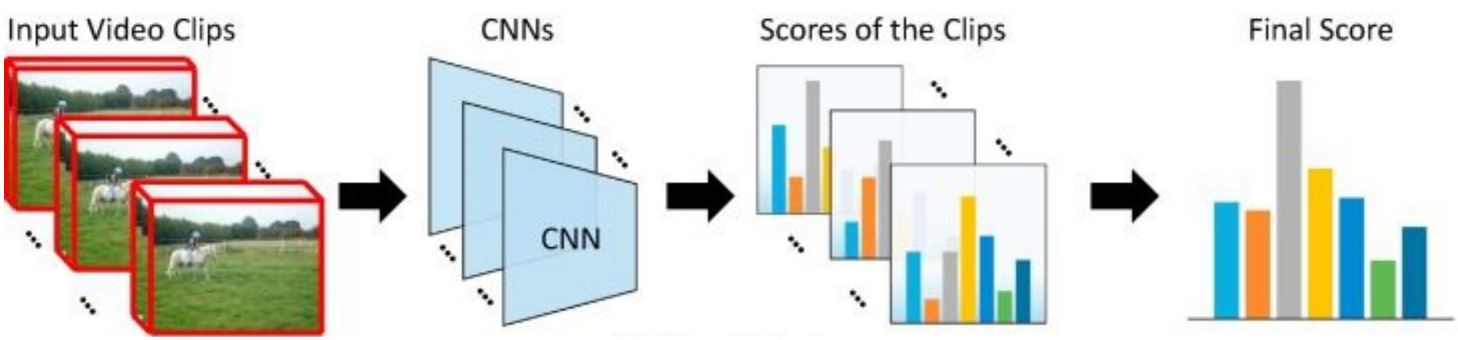
- A instance a set of frames with a label

Action: HulaHoop

How DL is Used for Action Recognition

It works but have some limitations:

- Dismiss temporal sense
- Dismiss object or human motion
- Don’t use multimodal information
- It is not an specialized network for video



How to introduce temporal information

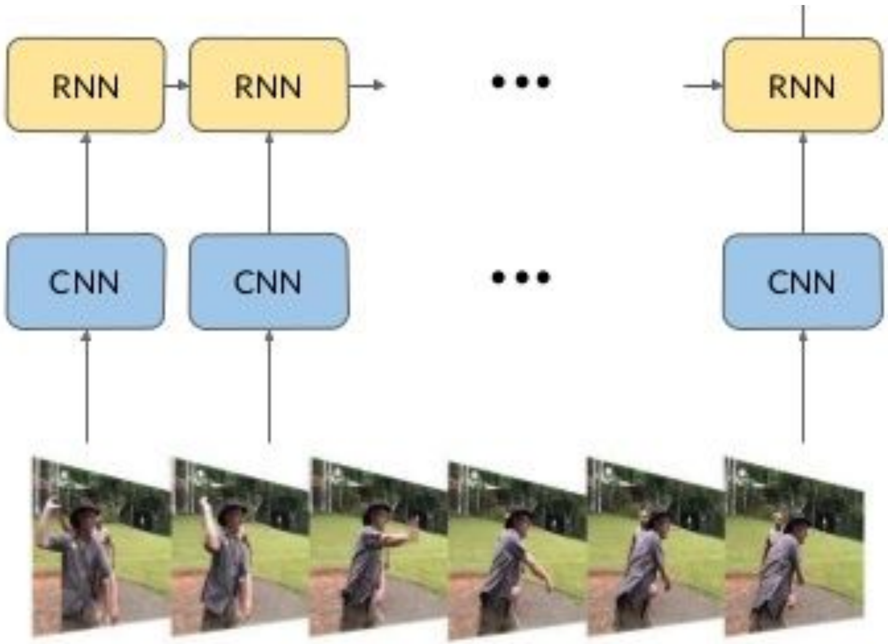
Temporal sense is important to correctly detect an action

- A video could be treated as a sequence
- Long-range dependency is desirable

How to introduce temporal information

The solution: 2D CNN + RNN

- RNN is suited for processing sequences
- This approach works better than 2D CNN.
- Problem: RNN cannot be parallelized



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